

Data Analytics Project 1

Data Cleaning & Preparation

Data Cleaning & Preparation

Objective

The objective of this project is to clean and prepare a raw dataset by identifying missing values, checking duplicate records, and validating data formats.

Tools Used

Microsoft Excel

Google Docs

Google Drive

Steps Performed

1. Missing Value Handling

The dataset was examined for missing values. Blank cells were identified in the CouponCode column and replaced with "No Coupon" to ensure data completeness.

Before:

Clipboard															Font															Alignment																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
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After:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	ORD2001	#####	C33540	Chair	1	273.19	275	Main	Debit Ca	Returnec	TRK6276	5	SAVE10	Faceboo	273.19
6	ORD2001	#####	C81840	Printer	4	626.01	668	Main	Online	Deliverex	TRK2924	8	SAVE10	Email	2504.04
7	ORD2001	#####	C37249	Phone	2	245.86	934	Main	Credit Ca	Shipped	TRK7293	4	SAVE10	Instagrar	491.72
8	ORD2001	#####	C83492	Laptop	1	664.42	986	Main	Gift Carc	Returnec	TRK9647	6	SAVE10	Faceboo	664.42
9	ORD2001	#####	C41460	Monitor	5	149.55	706	Main	Cash	Shipped	TRK7889	9	FREESH	Faceboo	747.75
10	ORD2001	#####	C26817	Phone	2	134.28	904	Main	Gift Carc	Cancelle	TRK6104	2	No Coup	Email	268.56
11	ORD2001	#####	C31946	Desk	4	509.38	102	Main	Credit Ca	Shipped	TRK3344	6	SAVE10	Google	2037.52
12	ORD2001	#####	C43443	Tablet	5	625.97	333	Main	Credit Ca	Returnec	TRK9887	9	WINTER	Instagrar	3129.85
13	ORD2001	#####	C39861	Monitor	3	49.14	831	Main	Online	Returnec	TRK4827	7	SAVE10	Email	147.42
14	ORD2001	#####	C38785	Monitor	2	180.5	179	Main	Debit Ca	Pending	TRK2047	6	FREESH	Referral	361
15	ORD2001	#####	C88348	Laptop	3	201.49	903	Main	Credit Ca	Returnec	TRK8191	4	FREESH	Faceboo	604.47
16	ORD2001	#####	C38474	Tablet	2	393.33	980	Main	Debit Ca	Pending	TRK7916	3	SAVE10	Instagrar	786.66
17	ORD2001	#####	C39416	Printer	1	473.96	942	Main	Cash	Deliverex	TRK5493	3	No Coup	Google	473.96
18	ORD2001	#####	C28976	Printer	2	533.81	914	Main	Credit Ca	Pending	TRK1319	3	FREESH	Email	1067.62
19	ORD2001	#####	C60672	Tablet	1	423.4	382	Main	Debit Ca	Shipped	TRK1197	3	No Coup	Referral	423.4
20	ORD2001	#####	C84329	Desk	1	431.44	891	Main	Gift Carc	Pending	TRK6515	3	SAVE10	Faceboo	431.44
21	ORD2001	#####	C14552	Monitor	5	224	273	Main	Cash	Returnec	TRK5910	9	SAVE10	Google	1120
22	ORD2001	#####	C54829	Chair	5	62.51	587	Main	Cash	Pending	TRK1671	9	WINTER	Instagrar	312.55
23	ORD2001	#####	C99023	Monitor	4	342.95	569	Main	Cash	Deliverex	TRK5264	6	No Coup	Google	1371.8
24	ORD2001	#####	C34416	Monitor	2	49.62	633	Main	Online	Pending	TRK7593	2	SAVE10	Google	99.24
25	ORD2001	#####	C42372	Phone	4	273.25	755	Main	Debit Ca	Cancelle	TRK3616	5	FREESH	Referral	1093
26	ORD2001	#####	C17470	Laptop	2	230.95	956	Main	Online	Deliverex	TRK2877	3	WINTER	Referral	461.9
27	ORD2001	#####	C42577	Monitor	1	399.98	709	Main	Debit Ca	Deliverex	TRK6943	3	WINTER	Email	399.98
28	ORD2001	#####	C21944	Laptop	5	59.55	296	Main	Online	Deliverex	TRK8027	6	No Coup	Referral	297.75
29	ORD2001	#####	C37593	Printer	1	472.74	881	Main	Online	Shipped	TRK8911	6	WINTER	Google	472.74
30	ORD2001	#####	C59823	Phone	4	160.07	568	Main	Online	Pending	TRK6585	5	No Coup	Email	640.28
31	ORD2001	#####	C33260	Desk	3	423.53	930	Main	Credit Ca	Deliverex	TRK5974	6	SAVE10	Referral	1270.59
32	ORD2001	#####	C44267	Monitor	2	426.7	895	Main	Online	Pending	TRK5475	5	SAVE10	Referral	853.4
33	ORD2001	#####	C26445	Chair	3	500.08	996	Main	Debit Ca	Returnec	TRK3122	6	No Coup	Faceboo	1500.24
34	ORD2001	#####	C12388	Tablet	5	536.72	830	Main	Cash	Deliverex	TRK9926	10	No Coup	Email	2683.6
35	ORD2001	#####	C52184	Chair	1	162.77	912	Main	Online	Cancelle	TRK7733	4	WINTER	Faceboo	162.77
36	ORD2001	#####	C67195	Chair	4	576.52	926	Main	Credit Ca	Pending	TRK5546	5	WINTER	Faceboo	2306.08
37	ORD2001	#####	C50304	Monitor	2	465.15	128	Main	Online	Shipped	TRK4688	2	SAVE10	Faceboo	930.3
38	ORD2001	#####	C59337	Laptop	2	534.43	465	Main	Debit Ca	Pending	TRK5096	5	No Coup	Faceboo	1068.86
39	ORD2001	#####	C10054	Laptop	2	76.74	409	Main	Cash	Returnec	TRK9401	7	SAVE10	Instagrar	153.48
40	ORD2001	#####	C15924	Laptop	1	245.89	549	Main	Debit Ca	Deliverex	TRK1483	3	SAVE10	Google	245.89
41	ORD2001	#####	C96664	Printer	3	256.46	666	Main	Credit Ca	Deliverex	TRK8556	4	SAVE10	Referral	769.38
42	ORD2001	#####	C84766	Desk	2	344.02	267	Main	Online	Deliverex	TRK7896	4	FREESH	Instagrar	688.04
43	ORD2001	#####	C20212	Monitor	3	611.45	657	Main	Gift Carc	Pending	TRK6977	4	FREESH	Instagrar	1834.35
44	ORD2001	#####	C28712	Printer	2	151.66	980	Main	Cash	Cancelle	TRK4077	6	No Coup	Instagrar	303.32
45	ORD2001	#####	C60572	Chair	3	499.21	871	Main	Online	Cancelle	TRK2038	8	No Coup	Instagrar	1497.63
46	ORD2001	#####	C73225	Phone	2	550.18	996	Main	Gift Carc	Returnec	TRK7384	3	FREESH	Referral	1100.36
47	ORD2001	#####	C17388	Chair	1	317.25	157	Main	Debit Ca	Returnec	TRK4701	1	SAVE10	Faceboo	317.25
48	ORD2001	#####	C43212	Monitor	4	550.14	223	Main	Debit Ca	Pending	TRK2510	4	FREESH	Instagrar	2200.56
49	ORD2001	#####	C38134	Desk	2	209.94	323	Main	Credit Ca	Returnec	TRK4248	3	FREESH	Email	419.88
50	ORD2001	#####	C12408	Monitor	2	80.77	464	Main	Debit Ca	Shipped	TRK3893	4	No Coup	Google	161.54
51	ORD2001	#####	C11985	Printer	1	30.44	633	Main	Credit Ca	Shipped	TRK5327	1	SAVE10	Email	30.44
52	ORD2001	#####	C36494	Tablet	1	179.22	446	Main	Online	Cancelle	TRK1467	6	FREESH	Instagrar	179.22

2. Duplicate Record Checking

The OrderID column was checked using Excel's Remove Duplicates feature. No duplicate records were found.

ORD200000															
OrderID	Date	Customer	Product	Quantity	Unit Price	Shipping	Address	Method	Order Status	Tracking Number	Items	Car Support	Col Referral	Source	Total Price
ORD2001	#####	C72649	Monitor	5	570.82	926	Main	Debit Ca	Shipped	TRK37947	7	SAVE10	Instagrar	2853.1	
ORD2001	#####	C75739	Phone	2	151.35	823	Main	Online	Shipped	TRK93186	3	SAVE10	Referral	302.7	
ORD2001	#####	C81728	Tablet	5	500.68	512	Main	Credit Ca	Cancelled	TRK42903	8	FREESH	Email	2753.4	
ORD2001	#####	C33540	Chair	1	273.19	275	Main	Debit Ca	Returnec	TRK6276	5	SAVE10	Faceboo	273.19	
ORD2001	#####	C81840	Printer	4	626.01	668	Main	Online	Delivered	TRK29241	8	SAVE10	Email	2504.04	
ORD2001	#####	C37249	Phone	2	245.86	934	Main	Credit Ca	Shipped	TRK72976	4	SAVE10			
ORD2001	#####	C83492	Laptop	1	664.42	986	Main	Gift Card	Returned	TRK96417	6	SAVE10			
ORD2001	#####	C41460	Monitor	5	149.55	706	Main	Cash	Shipped	TRK78809	9	FREESH			
ORD2001	#####	C26817	Phone	2	134.28	904	Main	Gift Card	Cancelled	TRK61042	2	No Coup			
ORD2001	#####	C31946	Desk	4	509.38	102	Main	Credit Ca	Shipped	TRK33478	6	SAVE10			
ORD2001	#####	C43443	Tablet	5	625.97	333	Main	Credit Ca	Returnec	TRK98859	9	WINTER			
ORD2001	#####	C39861	Monitor	3	49.14	831	Main	Online	Returned	TRK48234	7	SAVE10			
ORD2001	#####	C38785	Monitor	2	180.5	179	Main	Debit Ca	Pending	TRK20419	6	FREESH			
ORD2001	#####	C88348	Laptop	3	201.49	903	Main	Credit Ca	Returnec	TRK81513	4	FREESH			
ORD2001	#####	C38474	Tablet	2	393.33	980	Main	Debit Ca	Pending	TRK79186	3	SAVE10			
ORD2001	#####	C39416	Printer	1	473.96	942	Main	Cash	Delivered	TRK54930	3	No Coup			
ORD2001	#####	C28976	Printer	2	533.81	914	Main	Credit Ca	Pending	TRK13194	3	FREESH			
ORD2001	#####	C60672	Tablet	1	423.4	382	Main	Debit Ca	Shipped	TRK11972	3	No Coup	Referral	423.4	
ORD2001	#####	C84329	Desk	1	431.44	891	Main	Gift Card	Pending	TRK65151	3	SAVE10	Faceboo	431.44	
ORD2001	#####	C14552	Monitor	5	224	273	Main	Cash	Returnec	TRK59101	9	SAVE10	Google	1120	
ORD2001	#####	C54829	Chair	5	62.51	587	Main	Cash	Pending	TRK16718	9	WINTER	Instagrar	312.55	
ORD2001	#####	C99023	Monitor	4	342.95	569	Main	Cash	Delivered	TRK52646	6	No Coup	Google	1371.8	
ORD2001	#####	C34416	Monitor	2	49.62	633	Main	Online	Pending	TRK75937	2	SAVE10	Google	99.24	
ORD2001	#####	C42372	Phone	4	273.25	755	Main	Debit Ca	Cancelled	TRK36106	5	FREESH	Referral	1093	
ORD2001	#####	C17470	Laptop	2	230.95	956	Main	Online	Delivered	TRK28775	3	WINTER	Referral	461.9	
ORD2001	#####	C42577	Monitor	1	399.98	709	Main	Debit Ca	Delivered	TRK69423	3	WINTER	Email	399.98	
ORD2001	#####	C21944	Laptop	5	59.55	296	Main	Online	Delivered	TRK80276	6	No Coup	Referral	297.75	
ORD2001	#####	C37593	Printer	1	472.74	881	Main	Online	Shipped	TRK89117	6	WINTER	Google	472.74	
ORD2001	#####	C59823	Phone	4	160.07	568	Main	Online	Pending	TRK65895	5	No Coup	Email	640.28	
ORD2001	#####	C33260	Desk	3	423.53	930	Main	Credit Ca	Delivered	TRK59744	6	SAVE10	Referral	1270.59	
ORD2001	#####	C44267	Monitor	2	426.7	895	Main	Online	Pending	TRK54751	5	SAVE10	Referral	853.4	
ORD2001	#####	C26445	Chair	3	500.08	996	Main	Debit Ca	Returnec	TRK31229	6	No Coup	Faceboo	1500.24	
ORD2001	#####	C12388	Tablet	5	536.72	830	Main	Cash	Delivered	TRK99261	10	No Coup	Email	2683.6	
ORD2001	#####	C52184	Chair	1	162.77	912	Main	Online	Cancelled	TRK77312	4	WINTER	Faceboo	162.77	
ORD2001	#####	C67195	Chair	4	576.52	926	Main	Credit Ca	Pending	TRK55467	5	WINTER	Faceboo	2306.08	

3. Date Format Verification

Date columns were verified and standardized to maintain consistency across the dataset.

Get & Transform Data															Queries & Connections				Sort & Filter			Data Tools			
Get Data															Refresh All				Filter			Text to Columns			
From Text/CSV															Properties				Advanced			Flash Fill			
From Web															Workbook Links							Remove Duplicates			
From Table/Range																						Data Validation			
Recent Sources																						Consolidate			
Existing Connections																									
ORD000001															ORD000002				ORD000003			ORD000004			
ORD000005															ORD000006				ORD000007			ORD000008			
ORD000009															ORD000010				ORD000011			ORD000012			
ORD000013															ORD000014				ORD000015			ORD000016			
ORD000017															ORD000018				ORD000019			ORD000020			
ORD000021															ORD000022				ORD000023			ORD000024			
ORD000025															ORD000026				ORD000027			ORD000028			
ORD000029															ORD000030				ORD000031			ORD000032			
ORD000033															ORD000034				ORD000035			ORD000036			
ORD000037															ORD000038				ORD000039			ORD000040			
ORD000041															ORD000042				ORD000043			ORD000044			
ORD000045															ORD000046				ORD000047			ORD000048			
ORD000049															ORD000050				ORD000051			ORD000052			
ORD000053															ORD000054				ORD000055			ORD000056			
ORD000057															ORD000058				ORD000059			ORD000060			
ORD000061															ORD000062				ORD000063			ORD000064			
ORD000065															ORD000066				ORD000067			ORD000068			
ORD000069															ORD000070				ORD000071			ORD000072			
ORD000073															ORD000074				ORD000075			ORD000076			
ORD000077															ORD000078				ORD000079			ORD000080			
ORD000081															ORD000082				ORD000083			ORD000084			
ORD000085															ORD000086				ORD000087			ORD000088			
ORD000089															ORD000090				ORD000091			ORD000092			
ORD000093															ORD000094				ORD000095			ORD000096			
ORD000097															ORD000098				ORD000099			ORD000100			
ORD000101															ORD000102				ORD000103			ORD000104			
ORD000105															ORD000106				ORD000107			ORD000108			
ORD000109															ORD000110				ORD000111			ORD000112			
ORD000113															ORD000114				ORD000115			ORD000116			
ORD000117															ORD000118				ORD000119			ORD000120			
ORD000121															ORD000122				ORD000123			ORD000124			
ORD000125															ORD000126				ORD000127			ORD000128			
ORD000129															ORD000130				ORD000131			ORD000132			
ORD000133															ORD000134				ORD000135			ORD000136			
ORD000137															ORD000138				ORD000139			ORD000140			
ORD000141															ORD000142				ORD000143			ORD000144			
ORD000145															ORD000146				ORD000147			ORD000148			
ORD000149															ORD000150				ORD000151			ORD000152			
ORD000153															ORD000154				ORD000155			ORD000156			
ORD000157															ORD000158				ORD000159			ORD000160			
ORD000161															ORD000162				ORD000163			ORD000164			
ORD000165															ORD000166				ORD000167			ORD000168			
ORD000169															ORD000170				ORD000171			ORD000172			
ORD000173															ORD000174				ORD000175			ORD000176			
ORD000177															ORD000178				ORD000179			ORD000180			
ORD000181															ORD000182				ORD000183			ORD000184			
ORD000185															ORD000186				ORD000187			ORD000188			
ORD000189															ORD000190				ORD000191			ORD000192			
ORD000193															ORD000194				ORD000195			ORD000196			
ORD000197															ORD000198				ORD000199			ORD000200			
ORD000201															ORD000202				ORD000203			ORD000204			
ORD000205															ORD000206				ORD000207			ORD000208			
ORD000209															ORD000210				ORD000211			ORD000212			
ORD000213															ORD000214				ORD000215			ORD000216			
ORD000217															ORD000218				ORD000219			ORD000220			
ORD000221															ORD000222				ORD000223			ORD000224			
ORD000225															ORD000226				ORD000227			ORD000228			
ORD000229															ORD000230				ORD000231			ORD000232			
ORD000233																									

4. Data Validation

The dataset was reviewed to ensure:

- No missing values
- No duplicate IDs
- Consistent data formats
- Accurate records

Get Data

From Text/CSV

From Web

From Table/Range

Recent Sources

Existing Connections

Refresh All

Queries & Connections

Properties

Workbook Links

Sort

Filter

Clear

Reapply

Advanced

Text to Columns

Flash Fill

Remove Duplicates

Data Validation

Consolidate

Get & Transform Data

Sort & Filter

Data Tools

11

A

B

C

D

E

F

G

H

I

J

K

L

M

N

O

OrderID

Date

CustomerID

Product

Quantity

UnitPrice

ShippingAddress

PaymentMethod

OrderStatus

TrackingNumber

ItemsInCart

CouponCode

ReferralSource

TotalPrice

ORD2000C

04-01-2023

C72649

Monitor

5

570.62

928 Main

Debit Card

Shipped

TRK37947

7

SAVE10

Instagram

2853.1

ORD2000C

23-08-2024

C75739

Phone

2

151.35

823 Main

Online

Shipped

TRK91186

3

SAVE10

Referral

302.7

ORD2000C

27-02-2024

C81728

Tablet

5

550.68

512 Main

Credit Card

Cancelled

TRK42903

8

FREESHIP

Email

2753.4

ORD2000C

15-10-2023

C33540

Chair

1

273.19

275 Main

Debit Card

Returned

TRK62788

5

SAVE10

Facebook

273.19

ORD2000C

08-05-2025

C81840

Printer

4

626.01

668 Main

Online

Delivered

TRK29241

8

SAVE10

Email

2504.04

ORD2000C

23-10-2023

C37249

Phone

2

245.86

934 Main

Credit Card

Shipped

TRK72976

4

SAVE10

Instagram

491.72

ORD2000C

17-06-2025

C83492

Laptop

1

664.42

986 Main

Gift Card

Returned

TRK96417

6

SAVE10

Facebook

664.42

ORD2000C

12-05-2023

C41460

Monitor

5

149.55

706 Main

Cash

Shipped

TRK78809

9

FREESHIP

Facebook

747.75

ORD2000C

02-04-2025

C26817

Phone

2

134.28

904 Main

Gift Card

Cancelled

TRK61042

2

No Coupon

Email

268.56

ORD2000C

21-11-2023

C31946

Desk

4

509.38

102 Main

Credit Card

Shipped

TRK33478

6

SAVE10

Google

2037.52

ORD20001

29-12-2023

C43443

Tablet

5

625.97

333 Main

Credit Card

Returned

TRK98859

9

WINTER15

Instagram

3129.85

ORD20001

17-02-2024

C93861

Monitor

3

49.14

831 Main

Online

Returned

TRK48234

7

SAVE10

Email

147.42

ORD20001

15-10-2024

C38785

Monitor

2

180.5

179 Main

Debit Card

Pending

TRK20419

6

FREESHIP

Referral

361

ORD20001

30-08-2023

C88348

Laptop

3

201.49

903 Main

Credit Card

Returned

TRK81913

4

FREESHIP

Facebook

604.47

ORD20001

27-03-2023

C98474

Tablet

2

393.33

980 Main

Debit Card

Pending

TRK79186

3

SAVE10

Instagram

786.66

ORD20001

17-07-2023

C39416

Printer

1

473.96

942 Main

Cash

Delivered

TRK54930

3

No Coupon

Google

473.96

ORD20001

24-01-2023

C28976

Printer

2

533.81

914 Main

Credit Card

Pending

TRK13194

3

FREESHIP

Email

1067.62

ORD20001

02-03-2024

C60672

Tablet

1

423.4

382 Main

Debit Card

Shipped

TRK11972

3

No Coupon

Referral

423.4

ORD20001

26-03-2025

C84329

Desk

1

431.44

891 Main

Gift Card

Pending

TRK65191

3

SAVE10

Facebook

431.44

ORD20001

23-01-2023

C14552

Monitor

5

224

273 Main

Cash

Returned

TRK59101

9

SAVE10

Google

1120

ORD2000C

08-05-2023

C54829

Chair

5

62.51

587 Main

Cash

Pending

TRK16718

9

WINTER15

Instagram

312.55

ORD2000C

17-11-2024

C99023

Monitor

4

342.95

569 Main

Cash

Delivered

TRK52646

6

No Coupon

Google

1371.8

Result

The dataset was successfully cleaned and prepared for further analysis. All missing values were handled, duplicate records were checked, and data formats were validated.

Conclusion

Data cleaning is an essential step in the data analytics process. By ensuring data accuracy and consistency, the dataset becomes reliable for analysis and decision-making.

Cleaned excel

 **cleaned_Dataset for Data Analytics_project_1.xlsx**